

Particle Physics — Strong Interaction

IMAPP lecture notes

Spring 2026

1 Deep inelastic scattering

Deep inelastic scattering probes the internal structure of the proton at large momentum transfer. The relevant regime is characterized by

$$Q^2 = -q^2 > 0,$$

with Q^2 large compared with hadronic scales. At sufficiently high energy, the proton behaves as a collection of point-like constituents, the partons.

2 Quarks and colour

Quarks come in six flavours and carry colour charge. The colour degree of freedom is described by the fundamental representation of $SU(3)$, with three colour states. Observable hadrons are colour singlets:

$$3 \otimes \bar{3} = 1 \oplus 8, \quad 3 \otimes 3 \otimes 3 = 1 \oplus \dots$$

The absence of isolated coloured particles is confinement. The strong interaction is mediated by eight gluons, corresponding to the generators of $SU(3)$.

3 Parton distribution functions

The parton distribution function (PDF) $f_i(x, Q^2)$ gives the probability density to find a parton of type i carrying momentum fraction x at resolution Q^2 . A schematic factorization formula is

$$\sigma = \sum_i \int_0^1 dx f_i(x, Q^2) \hat{\sigma}_i(x, Q^2),$$

where $\hat{\sigma}_i$ is the hard partonic cross-section. PDFs are non-perturbative inputs and are measured or fitted from data.

4 Hadron colliders

At a collider such as the LHC, the partonic centre-of-mass energy is smaller than the proton-proton centre-of-mass energy because only fractions x_1 and x_2 of the proton momenta participate. The partonic energy satisfies $\hat{s} = x_1 x_2 s$. Increasing Q^2 resolves shorter distances and reveals more partonic structure.

5 Comparison with the weak interaction

The strong interaction is mediated by massless gluons and has a characteristic coupling that runs with the energy scale. The weak interaction is mediated by massive bosons, so its low-energy strength is suppressed by powers of $1/M_W^2$. Weak processes can change flavour and involve neutrinos; strong processes conserve flavour at the interaction vertex.

Summary

Deep inelastic scattering established the partonic structure of the proton. QCD describes quarks carrying colour and interacting through eight gluons, while PDFs connect the non-perturbative proton to calculable hard scattering.